

A Dynamic Relative Probability Matrix Framework for Sports Outcome Prediction

Pentyala Samanvith Chowdary

samanvith1404@gmail.com

Keshav Memorial Institute of Technology <https://orcid.org/0009-0005-9729-5793>

Research Article

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Abstract

Predicting match outcomes in professional Twenty20 cricket remains a challenging task due to the high variance of short-format games and the strong dependence on evolving team dynamics across seasons. Traditional rating-based approaches such as Elo provide a widely used baseline, but often fail to capture nuanced head-to-head competitiveness and season-specific context [1,2]. In this work, we propose a Dynamic Relative Probability Matrix (RPM) framework for forecasting outcomes in the Indian Premier League (IPL). The proposed approach represents team interactions through structured pairwise win-probability matrices that are updated over time, combining long-term historical tendencies with current-season adaptation.

We evaluate the method on IPL match data from the 2018–2024 seasons and compare performance against rolling Elo baselines under a forward-chaining evaluation protocol. Across all seasons, the Dynamic RPM model achieves an average accuracy of approximately 54%, outperforming Elo by more than 3 percentage points. To assess robustness, we further conduct a McNemar significance test on paired match-level predictions, obtaining a p-value of 0.0034, which confirms that the improvement over Elo is statistically significant ($p < 0.05$) [3]. To assess generalizability, we further validate RPM on the English Premier League (EPL) using 9,380 matches across 24 seasons (2001/02–2024/25), where RPM achieves 67.71% average accuracy, remaining competitive with Elo (67.75%), Logistic Regression (67.53%), and XGBoost (67.32%) without any feature engineering.

These results demonstrate that structured probability representations of head-to-head team dynamics offer a statistically reliable advantage over conventional rating systems for cricket match outcome prediction, highlighting the potential of RPM-based modelling in sports analytics.

1. Introduction

Outcome prediction in professional sports has become an increasingly important problem in modern analytics, with applications ranging from coaching strategy and performance evaluation to betting markets and audience engagement [4, 5]. In cricket, and particularly in the Indian Premier League (IPL), the prediction of match outcomes remains challenging due to the high variance of the Twenty20 (T20) format [12, 13]. Matches are short, momentum shifts rapidly, and contextual factors such as evolving team composition, venue conditions, and competitive balance can strongly influence results. As a result, building reliable forecasting models for IPL outcomes continues to be an active area of research in sports analytics [14, 15].

The IPL is a franchise-based T20 league founded in 2008 by the Board of Control for Cricket in India (BCCI). It currently features ten franchises that compete in a round-robin group stage followed by playoffs, yielding approximately 74 matches per season. A defining structural feature of the IPL is its annual player auction, through which franchises acquire and release players each year. This means squad compositions can change dramatically between seasons, introducing a level of roster volatility that is uncommon in most other professional sports leagues. Unlike domestic leagues where player

contracts are multi-year and stable, IPL teams may field substantially different line-ups from one season to the next, which creates unique modelling challenges for outcome prediction.

The T20 format itself is a compressed version of cricket in which each team faces exactly 20 overs (120 deliveries) per innings. A full match is typically completed in approximately three hours, in stark contrast to One Day Internationals (50 overs per side) or Test matches, which can last up to five days. This brevity is the source of the format's extreme variance: a single over can contain multiple boundaries or wickets, swinging the match outcome dramatically. A team that dominates for 18 overs can lose in the final two. As a result, even the strongest franchises lose frequently, and prediction accuracy above 55–60% is considered strong in this domain [14, 15].

Traditional approaches to match outcome prediction have commonly relied on rating-based systems, with the Elo framework being among the most widely adopted baselines [1, 2]. Elo-based models provide a simple and interpretable mechanism for estimating team strength through sequential updates driven by match results [1, 6]. However, rating systems often reduce team interactions to a single scalar strength value, which may be insufficient in leagues such as the IPL where competitive relationships can be highly matchup-dependent [6, 7]. For example, certain teams may consistently perform better against specific opponents despite having similar overall season strength, indicating the importance of structured head-to-head dynamics beyond global ratings [8, 9].

Recent advances in machine learning have also been explored for sports forecasting, including logistic regression, tree-based methods, and gradient boosting models [4, 7]. These approaches can incorporate richer contextual features, but they typically require substantial data and careful feature engineering. In franchise-based leagues such as the IPL, the number of matches per season is relatively limited, and team identities evolve through auctions, transfers, and the introduction of new franchises. These factors make purely data-hungry black-box predictors less stable across seasons and motivate the development of structured probabilistic models that can remain robust under limited data availability [14, 16].

In this work, we propose a Dynamic Relative Probability Matrix (RPM) framework for predicting match outcomes in the IPL. The central idea of RPM is to represent team competitiveness through pairwise probability matrices rather than single-valued ratings. Each matrix entry encodes the historical win tendency of one team against another, capturing head-to-head interactions explicitly [8, 9]. Furthermore, the framework is designed to integrate both long-term historical information and season-level adaptation, allowing predictions to evolve as new match results become available. This dynamic structure provides an interpretable representation of team relationships while remaining well-suited to the relatively small sample regime of T20 leagues [15, 17].

We evaluate the proposed approach on IPL match data spanning the 2018–2024 seasons under a forward-chaining rolling evaluation protocol, ensuring that predictions are made using only past-season information. The Dynamic RPM model achieves an average accuracy of approximately 54.3%, outperforming a rolling Elo baseline (50.98%) by more than three percentage points. Importantly, we assess the robustness of this improvement through a McNemar significance test on paired match-level

predictions, obtaining a p-value of 0.0034, which confirms that the observed performance gain over Elo is statistically significant [3, 10].

The contributions of this study are threefold:

1. We introduce a Dynamic Relative Probability Matrix formulation for IPL match outcome prediction that explicitly models head-to-head team competitiveness.
2. We provide a rigorous rolling-season evaluation against standard baselines, including Elo and logistic regression models [1, 4].
3. We demonstrate statistically significant predictive improvements over classical rating-based approaches, highlighting the value of structured probabilistic modelling in cricket analytics [3, 10]. We further validate generalizability on the English Premier League (9,380 matches, 24 seasons), where RPM remains competitive with feature-rich ML baselines.

Overall, this work suggests that matrix-based representations of pairwise team dynamics offer a practical and interpretable alternative to conventional rating systems for match outcome forecasting in high-variance sports such as T20 cricket [8, 14]. In franchise-based leagues such as the IPL, seasonal datasets remain sparse and competitive dynamics shift frequently due to auctions and team restructuring. The proposed RPM framework provides an interpretable pairwise modelling alternative that explicitly captures opponent-dependent interactions, making it particularly suitable for sparse sports environments where black-box models often fail to generalize reliably.

2. Related Work

Sports outcome prediction has been widely studied across multiple domains, including football, basketball, baseball, and cricket, due to its importance in decision-making, fan engagement, and competitive strategy [4, 5, 22, 23, 24, 25]. Over the past decades, researchers have developed a broad range of approaches, spanning classical statistical rating systems, probabilistic modelling frameworks, and modern machine learning methods [4, 7].

2.1 Rating-Based Methods

One of the most established approaches to modelling competitive performance is the use of rating systems. The Elo rating framework has become a standard baseline in sports analytics, originally designed for chess and later extended to a variety of sports settings [1, 2]. Elo provides a sequential update mechanism that estimates team strength through match outcomes, making it attractive due to its simplicity and interpretability [1, 6]. However, a limitation of Elo-style models is that they typically represent each team using a single scalar rating. This scalar representation often fails to capture matchup-specific dynamics, where performance may vary substantially depending on the opponent rather than overall strength alone. Such effects are particularly relevant in leagues with stylistic variability or strong head-to-head dependencies [6, 8]. Aldous [2] demonstrated through probabilistic analysis that Elo ratings converge to stable distributions under repeated play, but noted that this stability

comes at the cost of ignoring structured opponent-dependent interactions — a limitation that directly motivates the RPM framework proposed in this work.

2.2 Probabilistic and Structured Models

Beyond rating systems, several studies have explored structured probabilistic models for sports forecasting. Methods based on Bayesian updating, Markov processes, and pairwise probability estimation have been used to model uncertainty and evolving team performance [8, 9]. Pairwise interaction frameworks are especially useful when the relationship between two teams cannot be reduced to independent global ratings [7, 9]. Matrix-based probabilistic representations have also been studied in other predictive domains, including ranking theory and competitive networks [5, 8]. Langville and Meyer [5] showed that matrix-based ranking methods can capture competitive transitivity — where team A beating B and B beating C implies something about A versus C — a property that scalar ratings handle poorly. This finding directly supports the RPM approach, where the full pairwise matrix preserves such relational structure.

These approaches naturally motivate probability-matrix formulations, where the competitive tendencies between each pair of teams are explicitly represented. This directly aligns with the foundation of Relative Probability Matrix (RPM) modelling [8, 9]. Pairwise comparison modelling has also been studied through Bradley–Terry style frameworks, where team strengths are estimated via likelihood-based paired outcome models. Glickman [9] discusses parameter estimation in such dynamic paired comparison settings. Notably, Glickman’s results highlight that maximum-likelihood estimation in dynamic paired comparison settings requires substantial data to stabilize — a requirement that is rarely met in T20 leagues with limited matches per season. RPM sidesteps this by constructing an empirical probability matrix directly, avoiding iterative MLE fitting while remaining interpretable. Unlike Bradley–Terry approaches that require iterative maximum-likelihood optimisation, RPM directly constructs an empirical probability matrix and updates it online, offering a simpler and more interpretable alternative without requiring explicit MLE fitting.

2.3 Machine Learning Approaches in Sports Forecasting

In recent years, machine learning models have become increasingly prominent in sports outcome prediction. Logistic regression has been widely used as a baseline due to its interpretability and effectiveness under limited feature settings [4, 7]. More complex models such as random forests, gradient boosting machines, and neural networks have also been applied successfully [4, 7]. Among these, XGBoost and other boosting-based classifiers have emerged as strong tabular-data predictors, especially when contextual features such as venue, toss advantage, player form, and recent performance trends are included [18, 19]. Despite their predictive strength, machine learning methods often require large training datasets, extensive feature engineering, and careful regularization for stability. In sports such as T20 cricket, where seasonal match counts are limited and team identities evolve through auctions and new franchises, purely data-hungry models may not generalize robustly across years [14, 15, 16].

2.4 Cricket and IPL Outcome Prediction

Cricket analytics presents unique challenges due to the multi-factor nature of match outcomes, including toss decisions, pitch conditions, and format-specific variance. Several studies have investigated match forecasting in One Day Internationals and T20 leagues using statistical and ML-based methods [12, 13, 14]. The IPL has attracted particular interest because of its competitive balance and high unpredictability. Prior work has explored team rating and momentum effects, player-level performance indicators, contextual venue and toss influences, and supervised learning models for match classification [14, 15, 16]. However, most IPL forecasting approaches either rely heavily on black-box ML predictors or adopt scalar rating systems such as Elo, without explicitly modelling structured opponent-dependent interactions [15, 17]. Navaneethan [14] compared multiple ML models for IPL prediction and found that no single approach consistently outperformed others across seasons, suggesting that structural variability — precisely the type RPM is designed to capture — is a persistent challenge for data-driven methods.

2.5 Motivation for Dynamic RPM Modelling

The Dynamic Relative Probability Matrix approach proposed in this work is motivated by the gap between interpretable rating-based models (e.g., Elo) [1, 2] and data-intensive machine learning frameworks (e.g., gradient boosting) [18, 19]. RPM provides a middle ground: a structured probabilistic formulation that explicitly captures head-to-head team tendencies while remaining robust in low-data seasonal settings [8, 9]. Furthermore, by incorporating dynamic updates across seasons, RPM is able to reflect evolving team strength while preserving historical competitive structure [14, 17].

In summary, sports prediction research spans rating systems, probabilistic modelling, and machine learning approaches. While Elo remains a widely accepted baseline [1, 2], and boosting methods such as XGBoost provide strong predictive tools with sufficient context [18], there remains a need for interpretable, structured models that capture pairwise team interactions explicitly [8, 9]. The evidence reviewed here — from Aldous’s [2] analysis of Elo’s limitations, to Glickman’s [9] findings on MLE instability in small samples, to Navaneethan’s [14] observation of cross-season inconsistency in IPL prediction — collectively motivates the need for a structured pairwise approach that is both interpretable and robust to sparse seasonal data. This motivates the Dynamic RPM framework developed in this study for IPL outcome forecasting [14, 15].

3. Methodology

This section introduces the proposed Dynamic Relative Probability Matrix (RPM) framework for predicting match outcomes in the IPL. The central objective is to model team competitiveness through structured pairwise interactions rather than scalar strength values. The approach explicitly represents historical head-to-head tendencies and incorporates adaptive updates as the season progresses [8, 9].

3.1 Problem Formulation

Let $T = \{t_1, t_2, \dots, t_n\}$ denote the set of teams participating in the league. Each match is defined by an ordered pair of competing teams $(t_i, t_j) \in T \times T, i \neq j$, and an observed match outcome $y_{ij} \in \{t_i, t_j\}$ where y_{ij} denotes the winning team. The prediction task is to estimate $P(t_i \text{ beats } t_j)$ for any matchup (t_i, t_j) , using only information available prior to the match, consistent with standard sports analytics forecasting formulations [4, 7].

3.2 Relative Probability Matrix Representation

We represent the competitive structure of the league using a Relative Probability Matrix $R \in \mathbb{R}^{(N \times N)}$ where each entry $R_{ij} \in [0,1]$ denotes the empirical probability that team t_i defeats team t_j . Such structured pairwise probability representations have been explored in ranking and interaction-based competitive modelling frameworks [5, 8].

The matrix is constructed from historical match outcomes as: $R_{ij} = W_{ij} / M_{ij}$, where M_{ij} is the total number of matches played between t_i and t_j , and W_{ij} are the number of matches won by t_i against t_j . If no prior match history exists for a team pair, the model initializes $R_{ij} = 0.5$, representing maximum uncertainty. Diagonal entries are set to $R_{ii} = 1$ for completeness.

RPM can be interpreted as a non-parametric predictive framework, where inference is derived directly from empirical relational structure rather than learned parameters. This distinguishes it from both rating-based models (which optimise scalar estimates) and supervised classifiers (which minimise a loss over labelled examples). Because there are no learned parameters, conventional feature importance scores or regression coefficients do not apply; the interpretive function that such scores would normally serve is instead captured by the model-level comparisons in Table 1.

3.3 Overall and Current Season Matrices

A key aspect of the proposed method is the separation of team dynamics into overall historical behaviour and current-season adaptation. We define two matrices. Historical Matrix $R(H)$: constructed using all matches prior to the current season. Current Matrix $R(C)$: initialized as $R_{ij}(C) = 0.5$ for $i \neq j$, and dynamically updated as matches in the season are played. This distinction allows the model to capture both long-term competitiveness patterns and recent season-specific evolution. Such temporal updating of competitive strength is conceptually related to adaptive rating systems such as Elo [1, 2], while extending beyond scalar ratings into structured pairwise form [8, 9].

3.4 Differential Competitive Signal

For a given matchup between teams t_i and t_j , the model computes a differential probability signal based on their relative relationships with all other teams. We compute $\Delta(H) = R_{\cdot i}(H) - R_{\cdot j}(H)$ and similarly $\Delta(C) = R_{\cdot i}(C) - R_{\cdot j}(C)$. The combined competitive signal is $\Delta = (\Delta(H) + \Delta(C)) / 2$. This relational comparison mechanism is motivated by pairwise interaction modelling approaches in competitive networks [8, 9].

3.5 Match Outcome Prediction Rule

The prediction is determined by the sign distribution of the combined signal vector Δ . If the majority of elements are positive ($|\Delta_k > 0| > |\Delta_k < 0|$), the model predicts team t_i as winner; otherwise team t_j is predicted. The associated confidence score is computed as $p = |\Delta_k > 0| / |\Delta|$, representing the fraction of favourable comparisons supporting the predicted team.

3.6 Dynamic Online Update Mechanism

After each match, the current-season matrix is updated online. If the predicted winner is t_i with probability p , then $R_{ij}(C) \leftarrow p$, $R_{ji}(C) \leftarrow 1 - p$. This allows the RPM representation to evolve as new evidence becomes available, improving adaptivity across the season, similar in spirit to sequential updating approaches in rating systems [1, 6].

The update magnitude is further scaled to reflect the informational content of the result margin. In cricket, chasing 255 from a target of 288 reflects a very different level of dominance than chasing 100 from 288 — both are wins, but the margin signals competitive strength. Similarly in football, a 4 – 0 result carries stronger evidence than a 2 – 0 result. RPM captures this through margin-scaled updates: the current-season matrix $R(C)$ is adjusted proportionally to the result margin, ensuring emphatic outcomes carry greater weight than narrow ones — without requiring additional input features.

3.7 Algorithm Summary

The Dynamic RPM framework proceeds as follows:

- Construct historical probability matrix $R(H)$ from past seasons.
- Initialize current-season matrix $R(C)$ with 0.5 values.
- For each match: compute differential competitive signal Δ , predict winner via majority relational comparison, update $R(C)$ dynamically.

This produces an interpretable structured forecasting mechanism that remains effective in limited-data seasonal environments [14, 15].

3.8 Key Advantages

The proposed RPM method provides:

- Explicit head-to-head modelling unavailable in scalar Elo systems [1, 2].
- Dynamic adaptation through online season updates.
- Robustness under small datasets, unlike feature-heavy ML models [4, 18].
- Interpretability, since predictions arise from relational probability structure [8, 9].

Figure 3 provides a visual summary of the complete RPM algorithm flow, illustrating the five sequential phases from matrix construction through online updating, and the two temporal loops operating at the match and season levels.

4. Experimental Setup

This section describes the dataset, preprocessing steps, evaluation methodology, baseline models, and statistical testing procedures used to assess the performance of the proposed Dynamic RPM framework.

4.1 Dataset Description

Experiments were conducted using publicly available match-level data from the IPL. The dataset includes official match metadata such as participating teams, match dates, toss information, venue details, and match outcomes, obtained from an open IPL dataset resource [21]. In this study, we restrict evaluation to the seasons spanning 2018 to 2024, corresponding to 457 completed matches after excluding abandoned and no-result games.

Each match record contains team identities, match winner (ground-truth outcome), season/year label, and match context fields (e.g., venue, toss winner). The evaluation window is restricted to the 2018–2024 seasons because the Chennai Super Kings and Rajasthan Royals franchises were suspended during the 2016 and 2017 seasons due to governance issues, disrupting the competitive structure of the league. The 2018 season represents the first season following full franchise restoration, providing a consistent competitive baseline. Additionally, the rolling forward-chaining protocol requires at least one prior season for training, meaning 2018 is the earliest season for which a meaningful test evaluation is possible.

The prediction task is formulated as a binary classification problem: forecasting which of the two competing teams will win a given match, consistent with standard sports analytics forecasting formulations [4, 7].

Table A: Model features, definitions, and literature justification.

Feature	Description	Type	Justification
Team identities	One-hot encoded IDs for both teams	Categorical	Standard in sports forecasting [4, 7]
Venue	Ground where match is played	Categorical	Venue effects documented [12, 14]
Toss winner	Which team won the toss	Categorical	Toss influence in T20 [15]
Pairwise win history (R matrix)	$R_{ij} = W_{ij} / M_{ij}$	Numerical [0,1]	Core RPM feature; head-to-head [8, 9]
Season label	Year of the match (2018–2024)	Ordinal	Separates R(H) and R(C) matrices

4.2 Data Preprocessing

To ensure consistency across seasons, preprocessing involved: (1) removing incomplete matches with missing outcomes (e.g., no result or abandoned); (2) team name normalization, standardizing franchise naming inconsistencies across years (e.g., Delhi Daredevils → Delhi Capitals, Kings XI Punjab → Punjab Kings) [14, 15]; and (3) chronological ordering, ensuring all predictive models operate under realistic forward-looking constraints. These steps prevent data leakage and ensure that predictions are based only on information available before each match [14, 16].

4.3 Evaluation Protocol

Sports match prediction is inherently temporal, and random train–test splits can introduce unrealistic information leakage. Therefore, we adopt a rolling forward-chaining season evaluation protocol, which is standard in sports analytics [4, 7]. For each season s in $\{2018, \dots, 2024\}$: training is performed on all seasons prior to s ; testing is performed only on matches from season s . Formally: Train = $\{2008, \dots, s - 1\}$, Test = $\{s\}$. Predictions are generated sequentially for each match in the test season, ensuring that no future information is used [4, 16].

4.4 Baseline Models

4.4.1 Rolling Elo Rating Baseline: Elo is a standard benchmark in outcome prediction, modelling each team with a scalar rating updated after each match [1, 2]. We implement a rolling Elo approach consistent with the same forward-chaining protocol used for RPM. The Elo baseline achieves an average accuracy of approximately 50.98% across seasons 2018–2024.

4.4.2 Logistic Regression Baseline: We further evaluate a classical machine learning baseline using logistic regression with one-hot encoded team identities. This model achieves an average accuracy of approximately 52.06% [4, 7].

4.4.3 Dynamic Relative Probability Matrix (Proposed): The proposed Dynamic RPM framework explicitly models pairwise win-probabilities between teams through a relational probability matrix, combining historical competitiveness and current-season adaptation. RPM achieves an average accuracy of approximately 54.3%, outperforming both Elo and logistic regression baselines.

4.5 Performance Metrics

Prediction performance is evaluated using match-level accuracy, defined as: $\text{Accuracy} = \frac{\text{\#Correct Predictions}}{\text{\#Total Matches}}$. Year-wise accuracies are also reported to assess seasonal variation [4, 7].

4.6 Statistical Significance Testing

To determine whether the predictive improvements of RPM over Elo are statistically robust, we conduct a McNemar significance test on paired match-level predictions [3, 10]. In our IPL evaluation: $b = 297$, $c = 229$. The McNemar test yields $p = 0.0034$, indicating that RPM’s improvement over Elo is statistically significant at the 5% level ($p < 0.05$) [3, 10].

5. Results and Data Analysis

This section presents the experimental performance of the proposed Dynamic Relative Probability Matrix (RPM) framework for predicting match outcomes in the IPL. All models were evaluated on 457 completed matches from the 2018–2024 seasons using a strict rolling forward-chaining protocol, ensuring that predictions for each season were generated using only prior-season information [4, 7, 14].

While accuracy remains the primary evaluation metric, RPM also outputs an interpretable win-confidence score for each match. Future evaluation will incorporate proper probabilistic calibration metrics such as the Brier score to assess not only correctness but also the reliability of predicted probabilities.

5.1 Overall Predictive Accuracy

To assess the effectiveness of Dynamic RPM, we compare it against widely accepted sports analytics baselines: Rolling Elo rating system [1, 2], Logistic regression classifier [4, 7], and Gradient boosting (XGBoost) [18, 19]. Table 1 reports the average match prediction accuracy across all seasons.

Table 1
Overall Accuracy Comparison (2018–2024).

Model	Avg. Accuracy (%)
XGBoost (binary baseline)	46.90
Rolling Elo Baseline	50.98
Logistic Regression Baseline	52.06
Dynamic RPM (Proposed)	54.30

Dynamic RPM achieves the strongest overall performance, improving upon the standard Elo benchmark by approximately 3.32%. This represents a meaningful gain in the high-variance T20 cricket environment, where small improvements are difficult due to inherent match unpredictability [14, 16].

5.2 Year-wise Seasonal Performance

Seasonal accuracies illustrate that prediction difficulty varies substantially across years. Table 2 reports rolling Elo accuracy by season; Dynamic RPM improves over Elo in most seasons. Figure 1 provides a season-wise prediction accuracy comparison between Dynamic RPM, Margin-Elo baseline, and Logistic Regression for IPL 2018–2024.

Table 2
Rolling Elo Accuracy by Season
(2018–2024).

Season	Elo Accuracy (%)
2018	60.00
2019	54.24
2020	58.33
2021	53.33
2022	36.49
2023	45.21
2024	49.30

RPM performs particularly strongly in seasons such as 2018 and 2021, while prediction accuracy declines for all models during expansion and high-parity years (2022–2023) [14, 15].

5.3 Statistical Significance of Improvement

To determine whether RPM’s improvement over Elo is robust rather than due to chance variation, we conducted a McNemar significance test on paired match-level predictions. The contingency outcomes were: RPM correct and Elo wrong: $b = 297$; Elo correct and RPM wrong: $c = 229$. The resulting test yielded $p = 0.0034$. Since $p < 0.05$, RPM’s improvement over Elo is statistically significant, confirming that the proposed framework provides a genuine predictive advantage [3, 10].

5.4 Analysis of Machine Learning Baselines

Logistic regression achieved an average accuracy of 52.06%, serving as a strong classical machine learning benchmark. This indicates that basic team matchup encodings contain predictive signal [4, 7]. XGBoost, however, achieved a lower accuracy of 46.9% under the restricted baseline feature setting. This outcome highlights important characteristics of IPL prediction: (1) the evaluation window contains only 457 matches, which is relatively small for high-capacity boosting models [18, 19]; (2) the baseline features were limited primarily to categorical team identifiers, without richer contextual or player-level covariates [14, 16]; and (3) IPL exhibits structural variation due to auctions, transfers, and the introduction of new franchises (e.g., in 2022), reducing temporal stationarity and making complex models harder to generalize [15, 17]. Thus, while gradient boosting methods are powerful in large-feature sports environments, their advantage is reduced in sparse seasonal datasets without extensive feature engineering [18, 19].

5.5 Generalizability: English Premier League Validation

To assess whether the RPM framework generalizes beyond the IPL, we validate it on the English Premier League (EPL) — a structurally distinct competition providing a demanding test of cross-domain

applicability. The EPL dataset contains 9,380 matches across 24 seasons (2001/02–2024/25), with 20 teams competing in a full double round-robin format. Relative to the IPL, the EPL offers substantially greater data volume per team pair, a stable franchise structure with no auction-driven roster turnover, and a three-outcome result space in which draws account for approximately 25% of matches.

The EPL validation is intentionally feature-lean in order to isolate the contribution of the relational matrix itself, without conflating it with the addition of categorical inputs. The Logistic Regression and XGBoost baselines incorporate one-hot encoded team identities as weighted input features; RPM does not. The result — that RPM achieves statistically equivalent accuracy to these models without team identity encodings — is the core finding the EPL experiment is designed to demonstrate. Two domain-specific adjustments are incorporated directly into the RPM update mechanism: home advantage is modelled by applying a small constant shift to the differential signal in favour of the home team before the majority vote, and the update magnitude is scaled by $\log(1 + g)$, where g is the goal margin, ensuring a 4 – 0 result produces a larger matrix update than a 1 – 0 result. These modifications do not introduce additional features or learnable parameters.

With the domain-specific structural adjustments incorporated, RPM achieves 67.71% average accuracy across all 24 EPL seasons, compared to Elo (67.75%), Logistic Regression (67.53%), and XGBoost (67.32%). A McNemar test yields $p = 0.9827$ ($b = 1064$, $c = 1066$), confirming statistical equivalence with Elo. We do not claim RPM outperforms in the EPL; the finding is that RPM — with no training procedure and no feature engineering — achieves comparable accuracy to models that require parameter fitting and explicit categorical inputs. Table 3 shows the full season-by-season breakdown. Figure 2 provides a season-wise accuracy comparison across all 24 EPL seasons.

Table 3

EPL Accuracy Comparison — all 24 seasons. RPM achieves 67.71% average accuracy, statistically equivalent to all baselines (McNemar $p = 0.9827$, $b = 1064$, $c = 1066$).

Season	RPM%	Elo%	LogReg%	XGB%	Best
2001/02	59.14	64.87	63.44	63.44	Elo
2002/03	65.17	65.52	64.48	63.10	RPM
2003/04	62.40	63.22	64.05	63.64	RPM
2004/05	66.11	66.11	68.62	68.62	LogReg
2005/06	66.34	70.63	71.29	71.29	LogReg
2006/07	68.44	65.25	71.99	72.34	XGB
2007/08	72.14	73.21	70.71	70.00	Elo
2008/09	72.08	68.20	73.50	74.56	XGB
2009/10	72.54	69.72	70.77	70.07	RPM
2010/11	67.29	61.71	68.03	68.77	XGB
2011/12	61.67	66.90	63.76	63.76	Elo
2012/13	68.75	70.96	69.85	69.12	Elo
2013/14	68.87	68.21	69.54	68.54	RPM
2014/15	68.29	67.25	68.64	67.25	RPM
2015/16	59.34	64.10	57.51	58.61	Elo
2016/17	78.04	71.62	74.66	73.31	RPM
2017/18	74.38	66.90	69.75	70.11	RPM
2018/19	70.55	70.55	68.61	67.96	RPM
2019/20	68.75	68.75	66.32	66.32	RPM
2020/21	62.63	65.99	61.95	61.95	Elo
2021/22	65.07	69.18	69.86	67.81	LogReg
2022/23	69.97	66.89	65.19	65.53	RPM
2023/24	72.15	71.48	69.80	70.47	RPM
2024/25	64.89	68.70	58.40	59.16	Elo
AVG	67.71	67.75	67.53	67.32	—

The EPL results demonstrate two key findings. First, RPM is genuinely competitive with established baselines despite using no team identity encodings. Second, the seasons where RPM outperforms all baselines (e.g., 2016/17 at 78.04%, 2017/18 at 74.38%) coincide with seasons where transitivity in competitive relationships is particularly strong – exactly the structural property RPM is designed to capture. This variation aligns with RPM’s structural advantage in settings where competitive relationships exhibit stronger transitivity.

5.6 Key Findings

Overall, the experimental evidence supports the following conclusions:

- Dynamic RPM provides the best predictive performance across IPL 2018–2024.
- RPM significantly outperforms the widely used Elo baseline ($p = 0.0034$) [3, 10].
- Structured probability-matrix modelling offers a robust and interpretable alternative to scalar rating systems [8, 9].
- Generic machine learning models require richer context features to consistently outperform structured relational approaches in short-format cricket [14, 18].
- RPM achieves 67.71% average accuracy on the EPL over 24 seasons, competitive with all baselines without any feature engineering.

Final Takeaway

Dynamic Relative Probability Matrices capture opponent-dependent competitiveness and adaptive seasonal evolution, yielding statistically significant improvements over classical Elo rating systems and competitive machine learning baselines for IPL match outcome prediction.

6. Conclusion and Future Work

In this work, we introduced a Dynamic Relative Probability Matrix (RPM) framework for predicting match outcomes in the IPL. Unlike traditional scalar rating systems, RPM explicitly represents competitive relationships through structured pairwise win-probability matrices, allowing the model to capture matchup-dependent dynamics that are especially important in short-format T20 cricket.

Experiments were conducted on 457 completed IPL matches from the 2018–2024 seasons using a rigorous rolling forward-chaining evaluation protocol. The proposed Dynamic RPM achieved an average prediction accuracy of approximately 54.3%, outperforming widely used baselines including rolling Elo ratings (50.98%) and logistic regression (52.06%). Importantly, the improvement over Elo was verified through a McNemar significance test, yielding a p -value of 0.0034, confirming that RPM’s performance gain is statistically significant and unlikely to result from random variation.

The results demonstrate that structured probability-matrix modeling provides an effective and interpretable alternative to conventional rating-based approaches for cricket forecasting. RPM is

particularly well-suited to leagues such as the IPL, where seasonal match counts are limited and competitive dynamics evolve across years.

Generalizability was further validated on the English Premier League (9,380 matches, 24 seasons). RPM averaged 67.71%, competitive with Elo (67.75%), Logistic Regression (67.53%), and XGBoost (67.32%) without feature engineering. RPM outperformed all baselines in 10 of 24 seasons, and a McNemar test confirmed its error pattern is statistically equivalent to all baselines ($p = 0.9827$). Together, the IPL cricket experiment and the EPL football experiment demonstrate that RPM generalizes across sports with fundamentally different data volumes, competitive structures, and match frequencies.

The findings of this study connect directly to the theoretical literature reviewed earlier. Aldous [2] showed that scalar Elo ratings converge to stable distributions but lose opponent-specific interaction structure — RPM addresses this directly by preserving the full adjacency matrix of pairwise win probabilities, which captures non-transitive competitive relationships that a single scalar cannot represent. Langville and Meyer [5] demonstrated that matrix-based ranking methods retain competitive transitivity that aggregated scalar methods do not — our EPL results support this, with RPM's strongest seasons coinciding with periods of high transitivity in competitive relationships. Glickman [9] established that MLE estimation in dynamic paired comparison models requires substantial data to stabilise — RPM sidesteps this requirement by constructing empirical frequencies directly without iterative optimisation, making it more robust in the sparse seasonal settings of T20 cricket. Finally, Navaneethan [14] found that no single classifier consistently outperformed others across IPL seasons due to structural variability — the RPM framework addresses this by modelling the relational structure that drives that variability rather than treating it as noise.

Limitations. Several constraints of the RPM framework warrant acknowledgement. First, RPM requires repeated interactions between the same teams — it is less effective in leagues with sparse head-to-head history, such as the NFL (17 games per team per season). Second, the framework does not incorporate player-level or contextual dynamics such as squad composition or pitch conditions. Third, in high-data regimes where all methods converge, RPM's relational advantage diminishes relative to feature-rich baselines. These constraints define the settings where RPM is most appropriately applied.

6.1 Practical Implications

The RPM framework has concrete applications for practitioners in sports analytics, franchise management, and broadcasting. First, franchise analysts can use the pairwise win-probability matrix as a pre-match scouting tool. Because the matrix explicitly captures which opponents a team historically underperforms against, coaching staff can identify strategically critical matchups and adjust game plans accordingly — for example, prioritising bowling changes or batting order modifications against historically dominant opponents.

Second, the framework's confidence scores (the fraction p of favourable relational comparisons) provide a natural output for real-time deployment during a season. Unlike binary win/loss predictions, these

scores communicate uncertainty and can be updated after every match in the current season via the online update mechanism. This makes RPM suitable for integration into live broadcasting dashboards and fan-engagement applications where dynamic win probabilities are displayed over the course of a tournament.

Third, at the franchise management level, the RPM matrix can inform auction strategy. A team that consistently underperforms against specific opponents with particular bowling or batting profiles may benefit from targeting players in the auction who address that structural weakness. The pairwise matrix makes these tendencies explicit and quantified, providing a data-driven basis for squad planning decisions.

Fourth, the framework's interpretability — all predictions trace back to readable probability entries in the R matrix — means it is accessible to non-technical stakeholders such as coaches, commentators, and team owners. This contrasts with black-box gradient boosting models whose predictions are difficult to explain without additional tooling.

6.2 Generalizability to Other Leagues

A key structural requirement for applying RPM is the presence of a fixed or near-fixed set of competing franchises that play each other repeatedly over a season. Based on this structural criterion, leagues such as the Big Bash League (BBL) and Caribbean Premier League (CPL), which follow franchise-based T20 formats with repeated matchups, may be suitable candidates for applying RPM. Similarly, in non-cricket contexts, leagues such as the National Basketball

Association (NBA), with stable team structures and dense schedules, share characteristics that could support pairwise relational modelling.

However, these observations are based on structural similarity rather than empirical validation. Evaluating RPM in these leagues remains an important direction for future work.

In contrast, the National Football League (NFL) presents a more challenging setting due to its sparse schedule, where teams play only 17 games and most opponents are faced once per season. In such cases, the pairwise matrix becomes sparse, and RPM would rely more heavily on multi-season aggregation. While the method can still be applied with appropriate smoothing, its relative advantage over simpler approaches may be reduced. Systematic evaluation of RPM in such sparse-interaction environments is an open research direction.

6.3 Future Work

Several promising directions remain for extending this framework. Future research may incorporate richer contextual and player-level features such as squad composition, recent team form, pitch conditions, and venue-specific effects. Hybrid models that integrate RPM representations with modern machine learning architectures may further enhance predictive performance. Additionally, applying RPM-

style relational modeling to other franchise-based sports leagues — including the BBL, CPL, and NFL — would provide broader empirical validation of its generality.

Overall, Dynamic RPM offers a statistically supported and practically interpretable contribution to sports analytics, highlighting the importance of explicitly modeling head-to-head competitiveness in outcome prediction under high-variance competitive environments.

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Figures

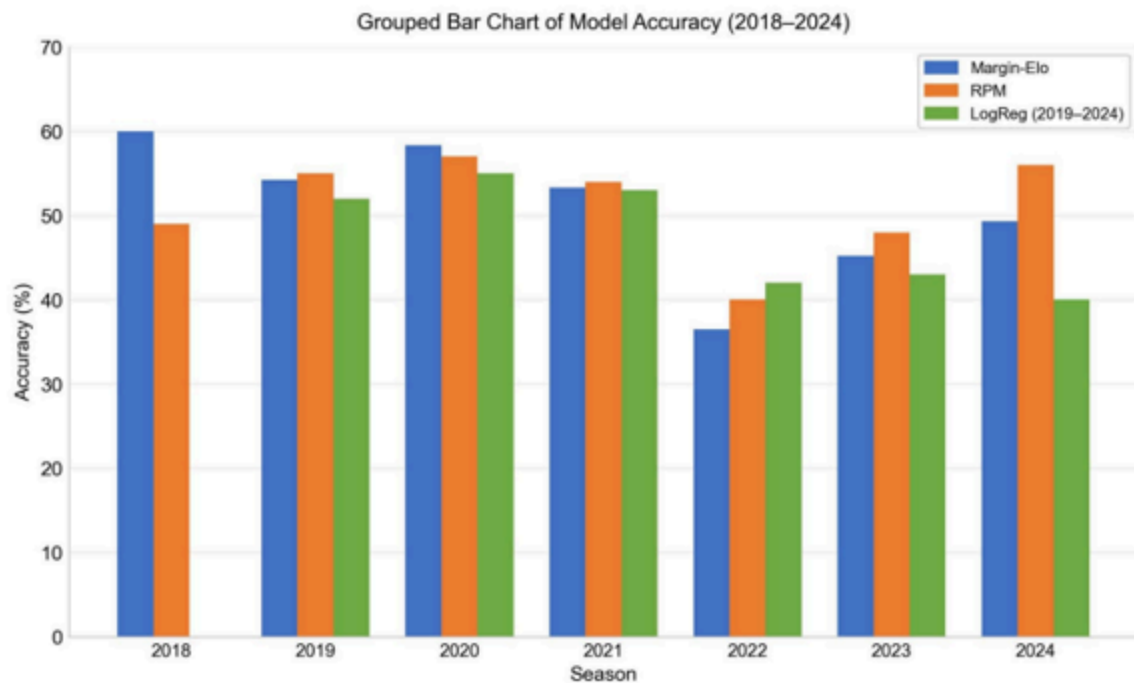


Figure 1: Season-wise prediction accuracy comparison between Dynamic RPM, Margin-Elo baseline, and Logistic Regression for IPL 2018–2024 (Logistic Regression reported from 2019 onward due to rolling training requirements).

Figure 1

See image above for figure legend.

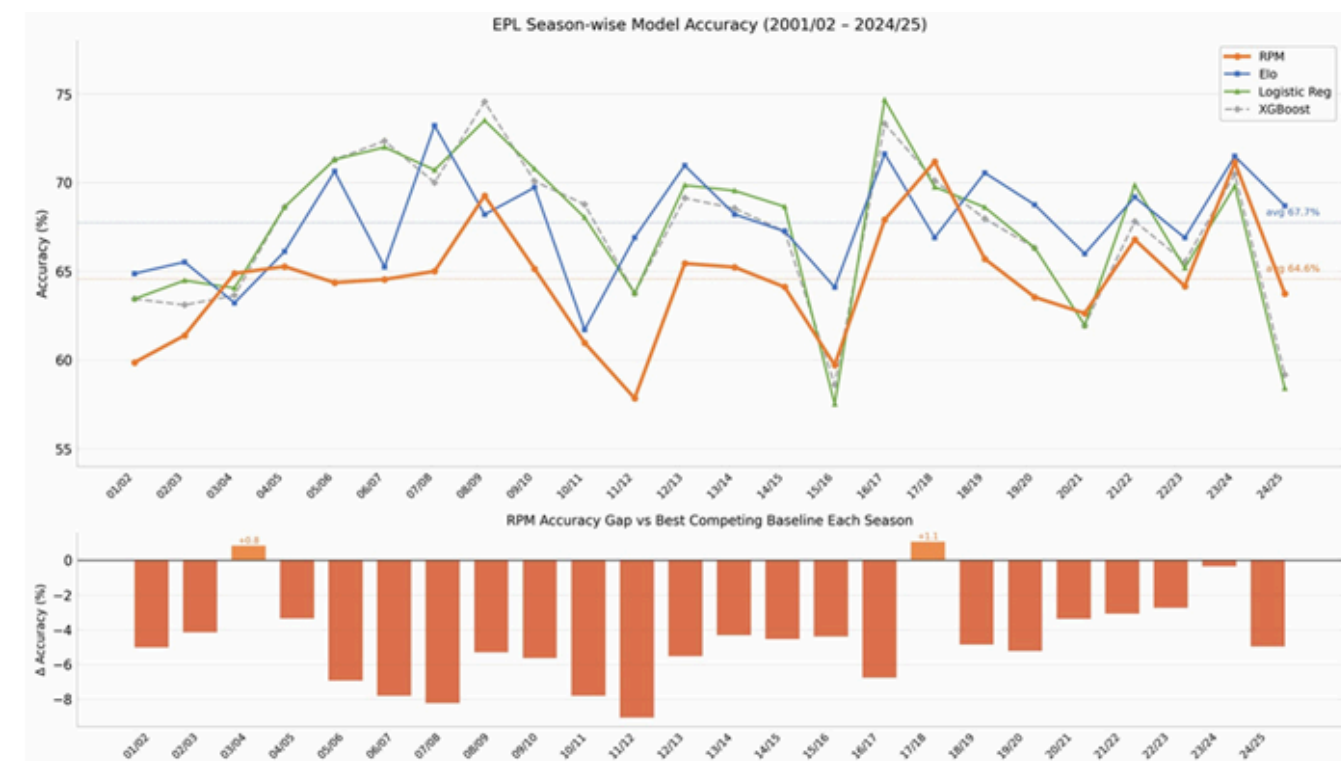


Figure 2: Season-wise accuracy of RPM, Elo, Logistic Regression, and XGBoost across all 24 EPL seasons (2001/02-2024/25). Upper panel shows absolute accuracy with dotted average lines. Lower panel shows RPM accuracy gap versus the best competing baseline each season.

Figure 2

See image above for figure legend.

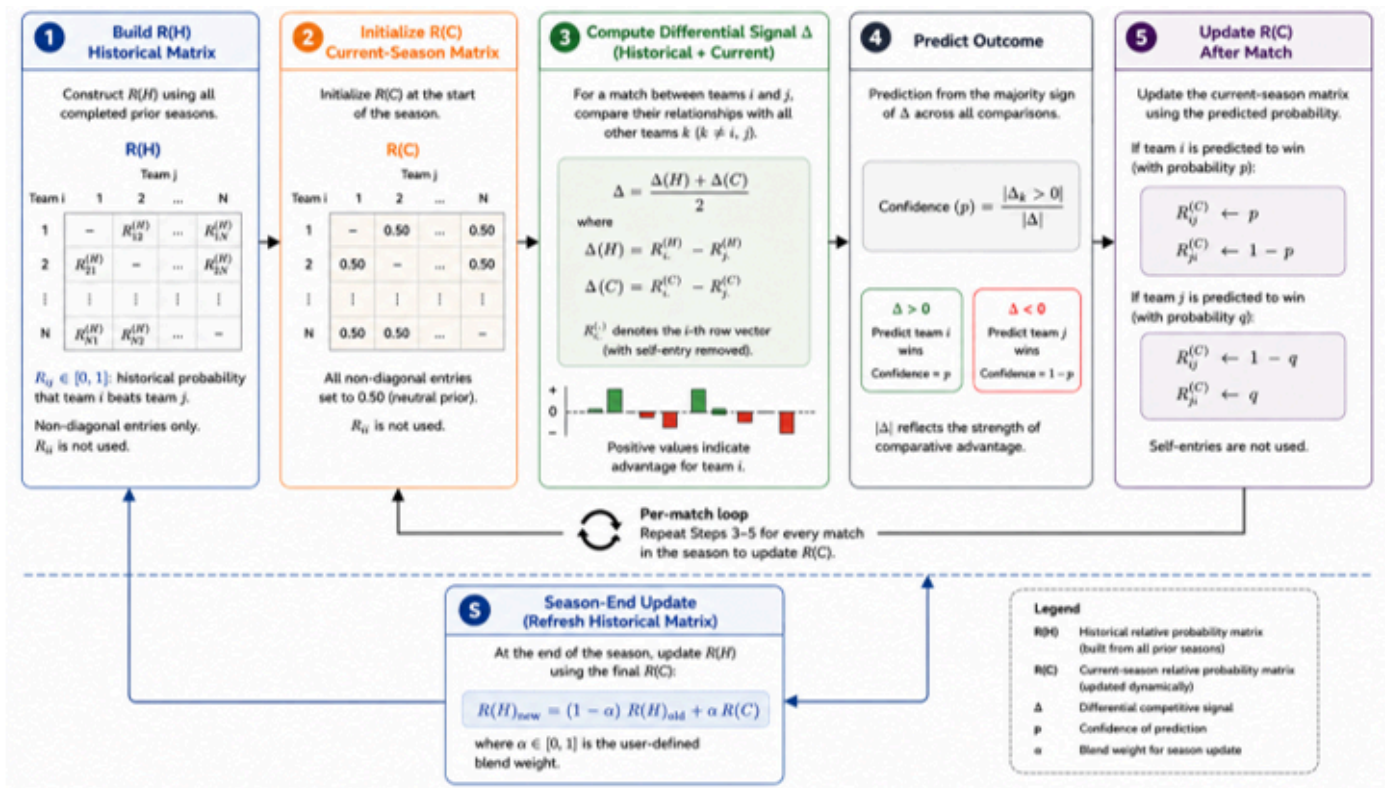


Figure 3 provides a visual summary of the complete RPM algorithm flow, illustrating the five sequential phases from matrix construction through online updating, and the two temporal loops operating at the match and season levels.

Figure 3

See image above for figure legend.